



DEVELOPERS LEVEL · SEMESTER 1

Homework Timer & Focus Tool

Student Workbook

JavaScript makes pages think and act — variables store, functions do, clicks trigger.

Name: _____

THEORY

Variables Store, Functions Do

JavaScript makes web pages interactive. A **variable** is a labelled box that stores a value (like the minutes on a timer). A **function** is a set of steps you can run by name. An **event handler** like `onclick` runs a function when the user clicks. `console.log` prints values so you can debug.

Discuss: what values would a homework timer need to store, and what should happen when you click Start?

Trace it

Read the code and answer: what is stored, and what runs on the click?

```
let minutes = 25;
```

```
function start() { console.log(minutes); }
```

```
<button onclick="start()">Start</button>
```

What value is stored in minutes? ____ What runs when the button is clicked? ____

Code we use:

let (variable) function onclick console.log if + -

Words we are learning

- **Variable** — a labelled box that stores a value: `let minutes = 25;`
- **Function** — named steps you can run again: `function start() { }`
- **Event handler** — `onclick` runs a function when a button is clicked.
- **console.log** — prints a value so you can check (debug) your code.
- **Focus** — the SEL skill this tool supports: managing your own time.

Name: _____ Date: _____

Variables Store Values

A **variable** stores a value. Match each variable to what it holds, then write your own.

let minutes = 25	the number of minutes
let name = "Sam"	a piece of text
let done = false	a true/false value

Write a variable that stores the number of seconds:

```
let _____ = _____;
```

Name: _____ Date: _____

Functions Do Steps

A **function** groups steps under a name. Number these steps 1-3 in a sensible order for `startTimer()`.

()	show the time on the page
()	set minutes to 25
()	count down each second

Write a name for a function that resets the timer:

```
function _____() { }
```

Name: _____ Date: _____

Clicks Trigger Functions

onclick runs a function when a button is clicked. Match each button to the function it should run.

Start button	start()
Pause button	pause()
Reset button	reset()

Circle the line that runs start() when clicked:

onclick="start()"

color="start"

Name: _____ **Date:** _____

Plan My Focus Timer

Plan your tool. Sketch the screen and list what it stores and does.



Sketch your timer screen

Variables I will use:

Functions I will use (what each does):

Name: _____ Date: _____

PROJECT

Homework Timer & Focus Tool — Plan

Build a timer that helps you focus: store the time in variables, run functions on clicks, and show a message when time is up.

1. What will my tool help me focus on?

2. Variables (name = value):

let _____ = _____;

let _____ = _____;

3. Functions and what each does:

4. Sketch the buttons and display:

buttons + timer display